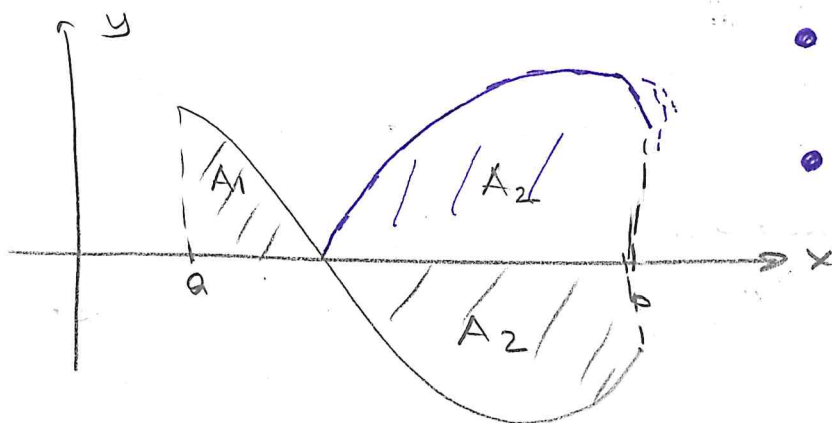


MATH 123 LECTURE NOTES

WEEK IX

AREAS OF PLANE REGIONS



$$\bullet \int_a^b f(x) dx = A_1 - A_2$$

$$\bullet \int_a^b |f(x)| dx = A_1 + A_2$$

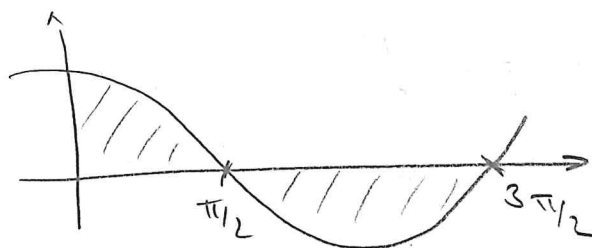
Ex Find the area bounded by $y = \cos x$, $y = 0$, $x = 0$

$$x = \frac{3\pi}{2}$$

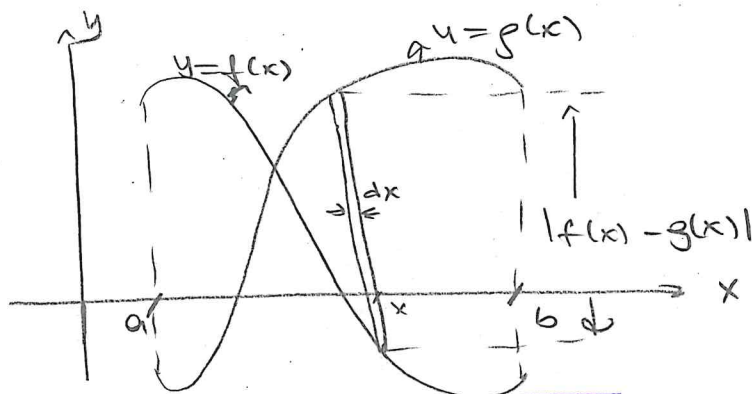
$$A = \int_0^{3\pi/2} |\cos x| dx$$

$$= \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{3\pi/2} (-\cos x) dx$$

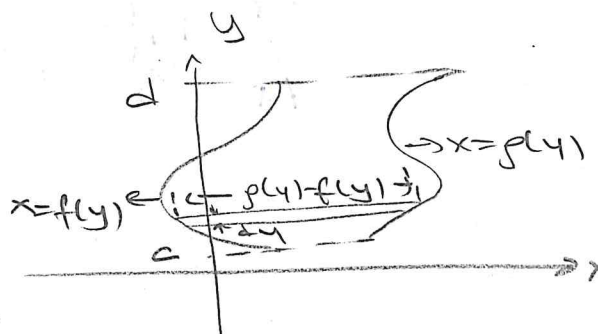
$$= \sin x \Big|_0^{\pi/2} - \sin x \Big|_{\pi/2}^{3\pi/2} = (1 - 0) - (-1 - 1) = 3.$$



AREAS BETWEEN CURVES

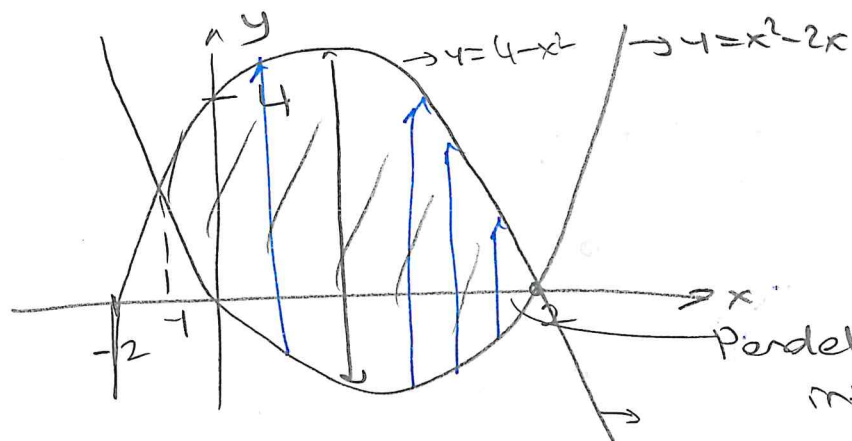


$$A = \int_a^b |f(x) - g(x)| dx$$



$$A = \int_c^d |g(y) - f(y)| dy$$

Ex 7 Find the area of the plane region R lying between the curves $y = x^2 - 2x$ and $y = 4 - x^2$



$$y = x^2 - 2x \quad \wedge \quad y = 4 - x^2$$

$$x^2 - 2x = 4 - x^2$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

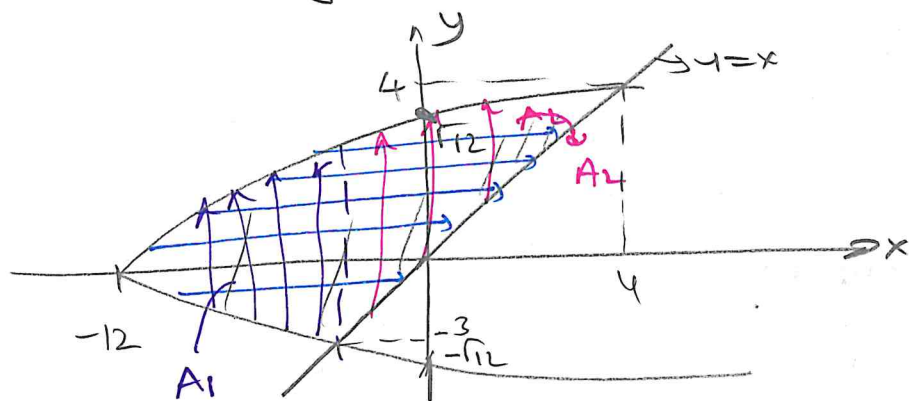
$$x = 2 \quad x = -1$$

Parallel's w.r.t. y axis
integral dx

$$\int_{-2}^2 ((4 - x^2) - (x^2 - 2x)) dx = \int_{-1}^2 (4 - 2x^2 + 2x) dx$$

$$= 4x - \frac{2x^3}{3} + x^2 \Big|_{-1}^2 = 4(2) - \frac{2}{3}(8) + 4 - (-4 + \frac{2}{3} + 1) = 9$$

Ex 7 Find the area of the plane region lying to the right of the parabola $x = y^2 - 12$ and to the left of the straight line $y = x$



$$y = x \quad \wedge \quad x = y^2 - 12$$

$$y = y^2 - 12$$

$$y^2 - y - 12 = 0$$

$$(y+3)(y-4) = 0$$

$$y = -3 \quad y = 4$$

I. WAY Parallel to x axis Integral dy

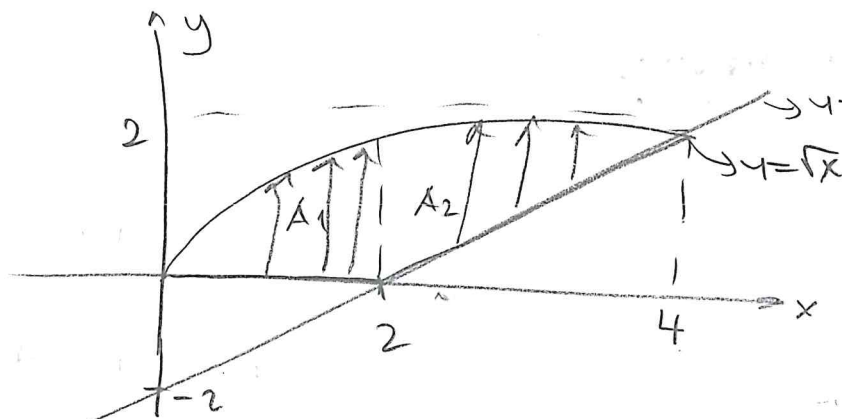
$$\int_{-3}^4 (y - (y^2 - 12)) dy = \left(\frac{y^2}{2} - \frac{y^3}{3} + 12y \right) \Big|_{-3}^4 = \frac{343}{6}$$

II. WAY Parallel to y axis, integral dx

$$\int_{-12}^{-3} (\sqrt{12+x} - (-\sqrt{12+x})) dx + \int_{-3}^4 (\sqrt{12+x} - x) dx$$

A_1 A_2

Ex Region in the first quadrant bounded by $y = \sqrt{x}$, $y = x - 2$ and the x -axis.



$$y = \sqrt{x} \quad \wedge \quad y = x - 2$$

$$\sqrt{x} = x - 2$$

$$x = x^2 - 4x + 4$$

$$x^2 - 5x + 4 = 0$$

$$x = 4 \quad x = 1$$

I. WAY $A = A_1 + A_2 = \int_0^2 (\sqrt{x} - 0) dx + \int_2^4 (\sqrt{x} - (x - 2)) dx$

Parallel to y axis
integral dx

A_1 A_2

$$= \left. \frac{x^{3/2}}{3/2} \right|_0^2 + \left(\left. \frac{x^{3/2}}{3/2} - \frac{x^2}{2} + 2x \right) \right|_2^4$$

II. WAY

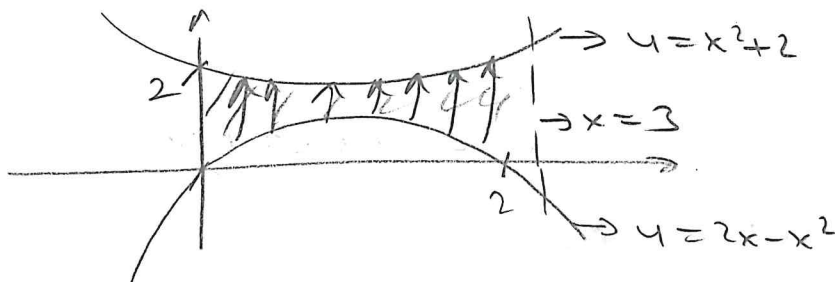
Parallel to
 x axis
integral dy

$$A = \int_{y=0}^{y=2} ((y+2) - y^2) dy = \left. \left(\frac{y^2}{2} + 2y - \frac{y^3}{3} \right) \right|_0^2$$

$$= 2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$$

$y = x - 2 \Rightarrow x = y + 2$
 $y = \sqrt{x} \Rightarrow x = y^2$

Ex Region in between $y = x^2 + 2$ and $y = 2x - x^2$, y axis and $x = 3$



$$A = \int_0^3 ((x^2 + 2) - (2x - x^2)) dx$$

$$= \int_0^3 (2x^2 - 2x + 2) dx$$

$$= \left. \left(\frac{2}{3} x^3 - x^2 + 2x \right) \right|_0^3$$

$$= 2 \cdot \frac{27}{3} - 9 + 6 = 15$$

So complicated for dy .

TECHNIQUES OF INTEGRATION

(I) INTEGRATION BY PARTS

→ Suppose $U(x)$ and $V(x)$ are two differentiable functions.

According to the product rule

$$\frac{d}{dx} (U(x) \cdot V(x)) = U(x) \frac{dV}{dx} + V(x) \frac{dU}{dx}$$

Integrating both sides of this equation and dropping terms we obtain

$$\int U(x) \frac{dV}{dx} dx = U(x)V(x) - \int V(x) \frac{dU}{dx} dx$$

or, more simply

$$\boxed{\int U dV = UV - \int V dU}$$

polynomial
LAP T's: $U = e^{\sin x}$
↓ $V = \cos x$ → power func.
logarithm trigonometry

Ex $\int x e^x dx = ?$

$$U = x \quad e^x dx = dV$$

$$dU = dx \quad e^x = V$$

$$\int x e^x dx = x \cdot e^x - \int dx \cdot e^x$$

$$\boxed{\int x e^x dx = x \cdot e^x - e^x + C}$$

$$\int U dV = U \cdot V - \int V dU$$

Ex $\int \ln x dx = ?$

$$\ln x = U \quad dx = dV$$

$$\frac{dx}{x} = dU \quad x = V$$

$$\int \ln x dx = \ln x \cdot x - \int x \frac{dx}{x}$$

$$\boxed{\int \ln x dx = \ln x \cdot x - x + C}$$

Ex $\int x^2 \sin x \, dx = ?$

$$\int x^2 \sin x \, dx = -\cos x \cdot x^2 + 2 \int \cos x \cdot x \, dx \quad u = x^2 \quad \sin x \, dx = dv$$

$$du = 2x \, dx \quad -\cos x = v$$

$$\int \cos x \cdot x \, dx = \sin x \cdot x - \int \sin x \, dx = \sin x \cdot x + \cos x + C$$

$$\begin{aligned} \cos x \, dx &= dv & x &= u \\ \sin x &= v & dx &= du \end{aligned}$$

$$\int x^2 \sin x \, dx = -x^2 \cos x + 2 [\sin x \cdot x + \cos x] + C$$

Ex $\int x \arctan x \, dx = ?$

$$\arctan x = u \quad x \, dx = dv$$

$$\frac{1}{1+x^2} \, dx = du \quad v = \frac{x^2}{2}$$

$$\begin{aligned} \int x \arctan x \, dx &= \frac{x^2}{2} \cdot \frac{1}{1+x^2} - \int \frac{x^2}{2(1+x^2)} \, dx \\ &= \frac{x^2}{2(1+x^2)} - \frac{1}{2} \int \frac{x^2+1-1}{1+x^2} \, dx \\ &= \frac{x^2}{2(1+x^2)} - \frac{1}{2} [x \arctan x] + C \\ &= \frac{x^2}{2(1+x^2)} - \frac{x}{2} + \frac{\arctan x}{2} + C \end{aligned}$$

Ex $\int \arcsin x \, dx = ?$

$$\arcsin x = u \quad dx = dv$$

$$\frac{1}{\sqrt{1-x^2}} \, dx = du \quad v = x$$

$$\begin{aligned} \int \arcsin x \, dx &= x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \, dx \\ &= x \arcsin x + \frac{1}{2} \frac{(1-x^2)^{1/2}}{1/2} \\ &= x \arcsin x + \sqrt{1-x^2} + C \end{aligned}$$

Ex 7 $I = \int \sec^3 x dx$

$$I = \int \sec^3 x dx$$

$$u = \sec x$$

$$dv = \sec^2 x dx$$

$$du = \sec x \cdot \tan x dx$$

$$v = \tan x$$

$$I = \int \sec^3 x dx = \sec x \cdot \tan x - \int \sec x \cdot \tan^2 x dx \quad \tan^2 x = \sec^2 x - 1$$

$$= \sec x \cdot \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \cdot \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$2 \int \sec^3 x dx = \sec x \cdot \tan x + \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x dx = \frac{\sec x \tan x + \ln |\sec x + \tan x| + C}{2}$$

Ex $I = \int e^{ax} \cos bx dx = ? \quad a \neq 0 \quad b \neq 0$

$$\int e^{ax} \cos bx dx = e^{ax} \cdot \frac{\sin bx}{b} - \frac{a}{b} \int \sin bx \cdot e^{ax} dx \quad u = e^{ax} \quad du = a \cdot e^{ax} dx$$

$$dv = \cos bx dx \quad v = \frac{\sin bx}{b}$$

$$\int \sin bx \cdot e^{ax} dx = -\frac{1}{b} e^{ax} \cos bx + \frac{a}{b} \int \cos bx \cdot e^{ax} dx \quad u = e^{ax} \quad du = a \cdot e^{ax} dx$$

$$v = -\frac{\cos bx}{b}$$

$$\int e^{ax} \cos bx dx = e^{ax} \frac{\sin bx}{b} - \frac{a}{b} \left[-\frac{1}{b} e^{ax} \cos bx + \frac{a}{b} \int \cos bx \cdot e^{ax} dx \right]$$

$$\left(1 + \frac{a^2}{b^2} \right) \int e^{ax} \cos bx dx = e^{ax} \cdot \frac{\sin bx}{b} + \frac{a}{b^2} e^{ax} \cos bx + C$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax} \cdot \frac{\sin bx}{b} + \frac{a}{b^2} e^{ax} \cos bx + C}{\left(1 + \frac{a^2}{b^2} \right)}$$

(b)

Ex 7 $I = \int_1^e x^3 (\ln x)^2 dx = ?$

$$u = (\ln x)^2$$

$$x^3 dx = dv$$

$$du = 2 \ln x \cdot \frac{1}{x} dx$$

$$\frac{x^4}{4} = v$$

$$\int_1^e x^3 \ln x = (\ln x)^2 \cdot \left(\frac{x^4}{4} \right) - \int 2 \ln x \cdot \frac{1}{x} \cdot \frac{x^4}{4} dx$$

$$= (\ln x)^2 \cdot \frac{x^4}{4} - \frac{1}{2} \int \ln x \cdot x^3 dx$$

$$\int \ln x \cdot x^3 dx = \ln x \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} dx$$

$$u = \ln x \quad x^3 dx = dv$$

$$du = \frac{1}{x} dx \quad \frac{x^4}{4} = v$$

$$= \ln x \cdot \frac{x^4}{4} - \frac{x^4}{16} + C$$

$$\int_1^e x^3 \ln x = (\ln x)^2 \cdot \frac{x^4}{4} - \frac{1}{2} \left[\ln x \cdot \frac{x^4}{4} - \frac{x^4}{16} \right] \Big|_1^e$$

$$= (\ln e)^2 \frac{e^4}{4} - \frac{1}{2} \left[\ln e \cdot \frac{e^4}{4} - \frac{e^4}{16} \right] - \left[(\ln 1)^2 \frac{1^4}{4} - \frac{1}{2} \left[\ln 1 \cdot \frac{1^4}{4} - \frac{1^4}{16} \right] \right]$$

$$= \frac{5e^4}{32} - \frac{1}{32}$$

II INTEGRALS OF RATIONAL FUNCTIONS

→ In this section we are concerned with integrals of the form

$$\int \frac{P(x)}{Q(x)} dx$$

where P and Q are polynomials.

Ex $\int \frac{x^3 + 3x^2}{x^2 + 1} dx = ?$

$\deg(P) = 3$ $\deg(Q) = 2$

$$\begin{array}{r} x^3 + 3x^2 \overline{) x^2 + 1} \\ x^2 + x \\ \hline 3x^2 - x \\ 3x^2 + 3 \\ \hline -x - 3 \end{array}$$

$$\int \frac{x^3 + 3x^2}{x^2 + 1} dx = \int (x+3) dx - \int \frac{(x+3)}{x^2 + 1} dx$$

$$= \int (x+3) dx - \int \frac{x}{x^2 + 1} dx - \int \frac{3}{x^2 + 1} dx$$

$$3x^2 - x$$

$$\frac{3x^2 + 3}{-x - 3}$$

$$\frac{x^3 + 3x^2}{x^2 + 1} = x + 3 - \frac{(x+3)}{x^2 + 1}$$

$$\int \frac{x^3 + 3x^2}{x^2 + 1} dx = \frac{x^2}{2} + 3x - \frac{1}{2} \ln|x^2 + 1| - 3 \arctan x + C$$

Ex $\int \frac{x}{2x-1} dx = ?$

$$\frac{x}{2x-1} = \frac{1}{2} \frac{2x}{2x-1} = \frac{1}{2} \frac{2x-1+1}{2x-1} = \frac{1}{2} \left(1 + \frac{1}{2x-1} \right)$$

$$\int \frac{x}{2x-1} dx = \frac{1}{2} \left[\int 1 dx + \int \frac{1}{2x-1} dx \right]$$

$$= \frac{1}{2} x + \frac{1}{2} \ln|2x-1| + C$$

The Basic Problem

Evaluate $\int \frac{P(x)}{Q(x)} dx$, where

the degree of $P <$ the degree of Q

① LINEAR AND QUADRATIC DENOMINATORS

Ⓐ LINEAR DENOMINATORS

→ Suppose that $Q(x)$ has degree 1. Thus, $Q(x) = ax+b$, where $a \neq 0$

Then, $P(x)$ must have degree 0 and be a constant c .

→ we have $\frac{P(x)}{Q(x)} = \frac{c}{ax+b}$ The substitution $u = ax+b$

leads to

$$\int \frac{c}{ax+b} dx = \frac{c}{a} \int \frac{du}{u} = \frac{c}{a} \ln|u| + C$$

• $c=1$ $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$

Ⓑ QUADRATIC DENOMINATORS

$$\int \frac{x}{x^2+a^2} dx, \int \frac{x}{x^2-a^2} dx, \int \frac{dx}{x^2+a^2}, \int \frac{dx}{x^2-a^2}$$

$$- \int \frac{x}{x^2+a^2} dx = \frac{1}{2} \ln(x^2+a^2) + C$$

$$- \int \frac{x}{x^2-a^2} dx = \frac{1}{2} \ln(x^2-a^2) + C$$

$$- \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$- \frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a} = \frac{A(x+a) + B(x-a)}{x^2-a^2}$$

$$A+B=0 \quad Aa-Ba=1 \quad \Rightarrow \quad A = \frac{1}{2a} \quad B = -\frac{1}{2a}$$

$$\begin{aligned} \int \frac{dx}{x^2-a^2} &= \frac{1}{2a} \int \frac{dx}{x-a} - \frac{1}{2a} \int \frac{dx}{x+a} \\ &= \frac{1}{2a} \ln|x-a| - \frac{1}{2a} \ln|x+a| + C \\ &= \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \end{aligned}$$

Partial Fractions

- Suppose $Q(x) = (x-a_1)(x-a_2)\dots(x-a_n)$ $a_i \neq a_j$ if $i \neq j$, $1 \leq i, j \leq n$
 If $P(x)$ is a polynomial of degree smaller than n , then
 $P(x)/Q(x)$ has a partial fraction decomposition of

$$\frac{P(x)}{Q(x)} = \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \dots + \frac{A_n}{x-a_n} \quad A_i \in \mathbb{R} \quad i=1,2,\dots,n$$

Ex $\int \frac{(x+4)}{x^2-5x+6} dx = ?$

$$\frac{x+4}{x^2-5x+6} = \frac{A}{x-3} + \frac{B}{x-2} = \frac{A(x-2) + B(x-3)}{x^2-5x+6} = \frac{(A+B)x - 2A - 3B}{x^2-5x+6}$$

$$\begin{aligned} A+B &= 1 \\ -2A-3B &= 4 \end{aligned} \quad \begin{aligned} B &= -6 \\ A &= 7 \end{aligned}$$

$$\begin{aligned} \int \frac{(x+4)}{x^2-5x+6} dx &= \int \frac{7}{x-3} dx + \int \frac{-6}{x-2} dx \\ &= 7 \ln|x-3| - 6 \ln|x-2| + C \end{aligned}$$

OR SHORTEST WAY $A = \frac{x+4}{x-2} \Big|_{x=3} = 7 \quad B = \frac{x+4}{x-3} \Big|_{x=2} = -6$
 (Erase $(x-3)$ and evaluate at 3)

Ex Evaluate $I = \int \frac{x^3+2}{x^3-x} dx = ?$

$$\begin{aligned} \int \frac{x^3+2}{x^3-x} dx &= \int \frac{x^3+2+x-x}{x^3-x} = \int \left(1 + \frac{x+2}{x^3-x}\right) dx \\ &= \int 1 dx + \int \frac{x+2}{x^3-x} dx = x + \int \frac{x+2}{x^3-x} dx \end{aligned}$$

$$\begin{aligned} x^3-x &= x(x^2-1) = x(x-1)(x+1) \\ \frac{x+2}{x^3-x} &= \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} \quad \text{OR} \quad A = \frac{x+2}{(x-1)(x+1)} \Big|_{x=0} = -2 \\ B &= \frac{x+2}{x(x+1)} \Big|_{x=1} = 3/2 \quad C = \frac{x+2}{x(x-1)} \Big|_{x=-1} = 1/2 \end{aligned}$$

$$\int \frac{x+2}{x^3-x} dx = \int \frac{-2}{x} dx + \int \frac{\frac{3}{2}}{x-1} dx + \int \frac{\frac{1}{2}}{x+1} dx$$

$$= -2 \ln|x| + \frac{3}{2} \ln|x-1| + \frac{1}{2} \ln|x+1| + C$$

$$\int \frac{x^3+2}{x^3-x} dx = x - 2 \ln|x| + \frac{3}{2} \ln|x-1| + \frac{1}{2} \ln|x+1| + C$$

Ex 7 $\int \frac{2+3x+x^2}{x(x^2+1)} dx$

$$\frac{2+3x+x^2}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{A(x^2+1) + (Bx+C)x}{x(x^2+1)}$$

$$A+B=1 \Rightarrow B=-1$$

$$C=3$$

$$A=2$$

$$\int \frac{2+3x+x^2}{x(x^2+1)} dx = \int \frac{2}{x} dx + \int \frac{-x+3}{x^2+1} dx$$

$x^2+1=u$
 $2x dx = du$

$$= 2 \ln|x| dx - \int \frac{x}{x^2+1} dx + \int \frac{3}{x^2+1} dx$$

$$= 2 \ln|x| - \frac{1}{2} \ln|x^2+1| + 3 \arctan x + C$$

Ex 7 $I = \int \frac{1}{x^3+1} dx = ?$

$$x^3+1 = (x+1)(x^2-x+1)$$

$$\frac{1}{x^3+1} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1} = \frac{A(x^2-x+1) + (Bx+C)(x+1)}{(x^3+1)}$$

$$A+B=0$$

$$-A+B+C=0$$

$$A+C=1$$

$$A = 1/3$$

$$B = -1/3$$

$$C = 2/3$$

$$\int \frac{dx}{x^3+1} = \frac{1}{3} \int \frac{dx}{x+1} - \frac{1}{3} \int \frac{x-2}{x^2-x+1} dx \quad x^2-x+1 = (x-\frac{1}{2})^2 + \frac{3}{4}$$

$$= \frac{1}{3} \ln|x+1| - \frac{1}{3} \int \frac{x - \frac{1}{2} - \frac{3}{2}}{(x-\frac{1}{2})^2 + \frac{3}{4}} dx \quad u = x - \frac{1}{2}$$

$$du = dx$$

$$= \frac{1}{3} \ln|x+1| - \frac{1}{3} \int \frac{u - \frac{3}{2}}{u^2 + \frac{3}{4}} du$$

$$= \frac{1}{3} \ln|x+1| - \frac{1}{3} \left[\int \frac{\overbrace{u}^{du/2}}{u^2 + \frac{3}{4}} du - \frac{3}{2} \int \frac{du}{u^2 + \frac{3}{4}} \right]$$

$$= \frac{1}{3} \ln|x+1| - \frac{1}{3} \left[\frac{1}{2} \ln|u^2 + \frac{3}{4}| - \frac{3}{2} \cdot \frac{2}{\sqrt{3}} \arctan\left(\frac{2u}{\sqrt{3}}\right) \right]$$

$$\int \frac{dx}{x^2+1} = \arctan x + C \quad \int \frac{dx}{x^2 + \frac{a^2}{b^2}} = \frac{b}{a} \arctan\left(\frac{b}{a} x\right)$$

$$= \frac{1}{3} \ln|x+1| - \frac{1}{6} \ln|x^2-x+1| + \frac{1}{\sqrt{3}} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + C$$

DENOMINATORS WITH REPEATED FACTORS

Ex $\int \frac{1}{x(x-1)^2} dx = ?$

$$\frac{1}{x(x-1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$= \frac{A(x^2-2x+1) + B(x^2-x) + Cx}{x(x-1)^2}$$

$$\begin{aligned} A+B &= 0 \\ -2A-B+C &= 0 \\ A &= 1 \end{aligned} \quad \begin{aligned} B &= -1 \\ C &= 1 \end{aligned}$$

$$\int \frac{1}{x(x-1)^2} dx = \int \frac{1}{x} dx - \int \frac{1}{x-1} dx + \int \frac{1}{(x-1)^2} dx$$

$$= \ln|x| - \ln|x-1| - \frac{1}{x-1} + C$$

Ex 7 $\int \frac{x^2+2}{4x^5+4x^3+x} dx = ?$

$$4x^5+4x^3+x = x(4x^4+4x^2+1) = x(2x^2+1)^2$$

$$\frac{x^2+2}{x(2x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{2x^2+1} + \frac{Dx+E}{(2x^2+1)^2}$$

$$= \frac{A(4x^4+4x^2+1) + B(2x^4+x^2) + C(2x^3+x) + Dx^2+Ex}{x(2x^2+1)^2}$$

$$4A+B=0 \quad 2C=0 \quad 4A+B+D=1 \quad C+E=0 \quad A=2$$

$$A=2 \quad B=-4 \quad C=0 \quad D=-3 \quad E=0$$

$$\int \frac{x^2+2}{4x^5+4x^3+x} dx = \int \frac{2}{x} dx + \int \frac{-4x}{(2x^2+1)} dx + \int \frac{-3x}{(2x^2+1)^2} dx$$

$2x^2+1=u$
 $4x dx = du$

$$= 2 \ln|x| - \ln|2x^2+1| + \frac{3}{4} \cdot \frac{1}{2x^2+1} + C$$

$$= \ln\left(\frac{x^2}{2x^2+1}\right) + \frac{3}{4} \cdot \frac{1}{2x^2+1} + C$$

THM Partial fraction Decomposition of rational function

Let P and Q be polynomials with real coefficients and suppose that the degree of P is less than degree of Q . Then,

$$f(x) = \frac{P(x)}{Q(x)}$$

(a) $Q(x)$ can be factored as follows

$$Q(x) = \underbrace{k}_{k \in \mathbb{R}} (x-a_1)^{m_1} (x-a_2)^{m_2} \dots (x-a_j)^{m_j} (x^2+b_1x+c_1)^{n_1} \dots (x^2+b_kx+c_k)^{n_k}$$

$x^2+b_ix+c_i$ is irreducible
No real roots

(b) degree $(Q) = m_1 + m_2 + \dots + m_j + n_1 + \dots + n_k$

(c) For each $(x-a_i)^{m_i}$ we have $\frac{A_1}{x-a_i} + \frac{A_2}{(x-a_i)^2} + \frac{A_3}{(x-a_i)^3} + \dots + \frac{A_{m_i}}{(x-a_i)^{m_i}}$

For each $(x^2+b_ix+c_i)^{n_i}$ we have $\frac{B_1x+C_1}{x^2+b_ix+c_i} + \frac{B_2x+C_2}{(x^2+b_ix+c_i)^2} + \dots + \frac{B_{n_i}x+C_{n_i}}{(x^2+b_ix+c_i)^{n_i}}$

(13)

INVERSE SUBSTITUTIONS

$$\int_a^b f(x) dx \xrightarrow{x=g(u)} \int_{x=a}^{x=b} f(g(u)) g'(u) du$$

The Inverse Trigonometric Substitutions

① $x = a \sin \theta$

Integrals involving $\sqrt{a^2 - x^2}$ ($a > 0$) can be reduced to a simpler form

$x = a \sin \theta$ or equivalently $\theta = \arcsin\left(\frac{x}{a}\right)$

Observe that $\sqrt{a^2 - x^2}$ makes sense only if $-a \leq x \leq a$, which corresponds to $-\pi/2 \leq \theta \leq \pi/2$. Since $\cos \theta \geq 0$ for this interval of θ , we have

$$\sqrt{a^2 - x^2} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta$$

Ex 7 $\int \frac{1}{(5-x^2)^{3/2}} dx = ?$

$$\int \frac{dx}{(5-x^2)^{3/2}} \quad \downarrow \quad \int \frac{\sqrt{5} \cos \theta}{(5 - 5 \sin^2 \theta)^{3/2}} d\theta$$

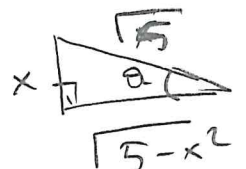
$$x = \sqrt{5} \sin \theta$$

$$dx = \sqrt{5} \cos \theta d\theta$$

$$= \int \frac{\sqrt{5} \cos \theta}{5^{3/2} (1 - \sin^2 \theta)^{3/2}} d\theta = \int \frac{\sqrt{5} \cos \theta}{5^{3/2} \cos^3 \theta} d\theta$$

$$= \frac{1}{5} \int \sec^2 \theta d\theta = \frac{1}{5} \tan \theta + C$$

$$= \frac{1}{5} \frac{x}{\sqrt{5-x^2}}$$



② $x = a \tan \theta$

Integrals involving $\sqrt{a^2 + x^2}$ or $\frac{1}{x^2 + a^2}$ (where $a > 0$) are often simplified by the substitution

$$\boxed{x = a \tan \theta \text{ or equivalently } \theta = \arctan\left(\frac{x}{a}\right)}$$

Since $x \in \mathbb{R}$, we have $-\pi/2 < \theta < \pi/2$ so $\sec \theta > 0$ and

$$\sqrt{a^2 + x^2} = a \sqrt{1 + \tan^2 \theta} = a \sec \theta$$

Ex $\int \frac{1}{(1+9x^2)^2} dx$

$$\int \frac{1}{(1+9x^2)^2} dx = \int \frac{\frac{\sec^2 \theta}{3} d\theta}{(1+\tan^2 \theta)^2} = \frac{1}{3} \int \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^2}$$

$3x = \tan \theta$
 $3dx = \sec^2 \theta d\theta$

$\frac{1}{\sec^2 \theta} = \cos^2 \theta$

$$= \frac{1}{3} \int \cos^2 \theta d\theta = \frac{1}{3} \int \frac{\cos 2\theta + 1}{2} d\theta$$

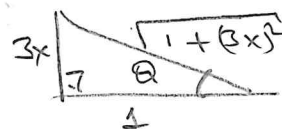
$$= \frac{1}{3} \cdot \frac{1}{2} \left(\frac{\sin 2\theta}{2} + \theta \right) + C$$

$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= \cos^2 \theta - (1 - \cos^2 \theta) \\ &= 2\cos^2 \theta - 1 \end{aligned}$$

$$= \frac{1}{6} \cdot \theta + \frac{1}{6} \cdot \frac{\sin 2\theta}{2} + C$$

$$\frac{\cos 2\theta + 1}{2} = \cos^2 \theta$$

$$= \frac{1}{6} \arctan(3x) + \frac{1}{6} \cdot \frac{3x}{\sqrt{1+9x^2}} \cdot \frac{1}{\sqrt{1+9x^2}} + C \quad \begin{aligned} \theta &= \arctan(3x) \\ \tan \theta &= 3x \end{aligned}$$



III

$$\underline{x = a \sec \theta}$$

Integrals involving $\sqrt{x^2 - a^2}$ (where $a > 0$) frequently be simplified by using the substitution.

$$\underline{x = a \sec \theta \text{ or equivalently } \theta = \operatorname{arcsec} \frac{x}{a}}$$

$$\sqrt{x^2 - a^2} = a \sqrt{\sec^2 \theta - 1} = a \sqrt{\tan^2 \theta} = a |\tan \theta|$$

Observe that $\sqrt{x^2 - a^2}$ makes sense for $x \geq a$ and for $x \leq -a$

If $x \geq a$, then $0 \leq \theta = \operatorname{arcsec} \left(\frac{x}{a} \right) = \arccos \left(\frac{a}{x} \right) < \pi/2$ and $\tan \theta \geq 0$

If $x \leq -a$, then $\pi/2 < \theta = \operatorname{arcsec} \left(\frac{x}{a} \right) = \arccos \left(\frac{a}{x} \right) \leq \pi$ and $\tan \theta \leq 0$

→ In the first case $\sqrt{x^2 - a^2} = a \tan \theta$, in the second case

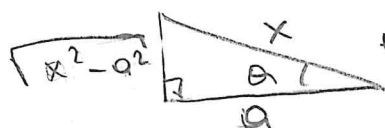
$$\sqrt{x^2 - a^2} = -a \tan \theta$$

Ex $\int \frac{dx}{\sqrt{x^2 - a^2}} = ?$ where $a > 0$

Assume $x \geq a$

$$x = a \sec \theta$$

$$dx = a \sec \theta \tan \theta d\theta$$



$$\tan \theta = \frac{\sqrt{x^2 - a^2}}{a}$$

$$\sqrt{x^2 - a^2} = a \tan \theta$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \int \frac{a \sec \theta \tan \theta d\theta}{a \tan \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C = \ln |x + \sqrt{x^2 - a^2}| - \ln a + C$$

Ex $\int \frac{dx}{\sqrt{x^2 - a^2}} = ? \quad a > 0$

Assume $x < -a$ $u = -x$ $u > a$ $du = -dx$

$$I = \int \frac{dx}{\sqrt{x^2 - a^2}} = - \int \frac{du}{\sqrt{u^2 - a^2}} = - \ln |u + \sqrt{u^2 - a^2}| + C_1$$

$$= \ln |u + \sqrt{u^2 - a^2}|^{-1} + C$$

$$= \ln \left| \frac{1}{-x + \sqrt{x^2 - a^2}} \right| + C$$

$$= \ln \left| \frac{1}{-x + \sqrt{x^2 - a^2}} \cdot \frac{x + \sqrt{x^2 - a^2}}{x + \sqrt{x^2 - a^2}} \right| + C$$

$$= \ln \left| \frac{x + \sqrt{x^2 - a^2}}{(x^2 - a^2) - x^2} \right| + C = \ln \left| \frac{x + \sqrt{x^2 - a^2}}{-a^2} \right| + C$$

$$= \ln |x + \sqrt{x^2 - a^2}| - \ln a^2 + C$$

Ex $\int \frac{1}{\sqrt{2x - x^2}} dx = ?$

$$\int \frac{1}{\sqrt{2x - x^2}} dx = \int \frac{dx}{\sqrt{1 - \underbrace{(1 - 2x + x^2)}_{(x-1)^2}}}$$

$$= \int \frac{dx}{\sqrt{1 - (x-1)^2}} \quad \begin{array}{l} u = x-1 \\ du = dx \end{array}$$

$$= \int \frac{du}{\sqrt{1 - u^2}} = \arcsin u + C = \arcsin(x-1) + C$$

Ex 7

$$\int \frac{1}{1+\sqrt{2x}} dx = \int \frac{u du}{1+u}$$

$$\sqrt{2x} = u$$

$$2x = u^2$$

$$2dx = 2u du$$

$$= \int \frac{u+1-1}{1+u} du = \int \left(1 - \frac{1}{u+1}\right) du$$

$$= u - \ln|u+1| + C$$

$$= \sqrt{2x} - \ln|1+\sqrt{2x}| + C$$

Ex 7

$$\int_{-1/3}^2 \frac{x}{\sqrt[3]{3x+2}} dx = ?$$

$$3x+2 = u^3$$

$$3dx = 3u^2 du$$

$$x=2 \Rightarrow u^3=8 \quad u=2$$

$$x=-1/3 \Rightarrow u^3=1 \quad u=1$$

$$= \int_{u=1}^{u=2} \frac{u^3-2}{3 \cdot u} \cdot u^2 du$$

$$= \frac{1}{3} \int_1^2 (u^4 - 2u) du = \frac{1}{3} \left(\frac{u^5}{5} - u^2 \right) \Big|_1^2 = \frac{16}{15}$$

$$\int \frac{1}{\sqrt{x}(1+x^{1/3})} dx = ?$$

$$\int \frac{1}{\sqrt{x}(1+x^{1/3})} dx = \int \frac{6u^5 du}{u^3(1+u^2)} = 6 \int \frac{u^2}{1+u^2} du$$

$$x = u^6$$

$$dx = 6u^5 du$$

$$= 6 \int \left(1 - \frac{1}{1+u^2}\right) du$$

$$= 6u - 6 \arctan u + C$$

$$= 6(x^{1/6} - \arctan x^{1/6} + C)$$

tan($\theta/2$) substitution

$$x = \tan \frac{\theta}{2} \Rightarrow \cos^2 \frac{\theta}{2} = \frac{1}{\sec^2 \theta/2} = \frac{1}{1 + \tan^2 \frac{\theta}{2}} = \frac{1}{1+x^2}$$

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1 = \frac{2}{1+x^2} - 1 = \frac{1-x^2}{1+x^2}$$

$$\sin \theta = 2 \cos \frac{\theta}{2} \cdot \sin \frac{\theta}{2} = 2 \cdot \tan \frac{\theta}{2} \cdot \cos^2 \frac{\theta}{2} = \frac{2x}{1+x^2}$$

$$dx = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$$

$$d\theta = 2 \cos^2 \frac{\theta}{2} dx = \frac{2 dx}{1+x^2}$$

SUMMARY

$$\cos \theta = \frac{1-x^2}{1+x^2}$$

$$\sin \theta = \frac{2x}{1+x^2}$$

$$d\theta = \frac{2 dx}{1+x^2}$$

Ex $\int \frac{1}{2+\cos \theta} d\theta$

$$x = \tan(\theta/2)$$

$$\cos \theta = \frac{1-x^2}{1+x^2}$$

$$d\theta = \frac{2 dx}{1+x^2}$$

$$= \int \frac{\frac{2}{1+x^2} dx}{2 + \frac{1-x^2}{1+x^2}} dx = 2 \int \frac{1}{3+x^2} dx$$

$$= \frac{2}{\sqrt{3}} \arctan\left(\frac{x}{\sqrt{3}}\right) + C$$

$$= \frac{2}{\sqrt{3}} \arctan\left(\frac{1}{\sqrt{3}} \tan \frac{\theta}{2}\right) + C$$

